

References

This page contains all references cited in the TSENAT package, organized by topic. For a complete list in BibTeX format, see the vignettes/TSENAT.bib file.

Information Theory & Entropy

Foundational Work

- **Shannon, C. E.** (1948). “A mathematical theory of communication.” *The Bell System Technical Journal*, 27(3-4), 379-423.
- **Simpson, E. H.** (1949). “Measurement of diversity.” *Nature*, 163, 688. doi: 10.1038/163688a0
- **Kullback, S., & Leibler, R. A.** (1951). “On information and sufficiency.” *Annals of Mathematical Statistics*, 22(1), 79-86. doi: 10.1214/aoms/1177729694

Tsallis Entropy Theory

- **Tsallis, C.** (2017). “On the foundations of statistical mechanics.” *European Physical Journal Special Topics*, 226, 1433-1443. doi: 10.1140/epjst/e2016-60252-2
- **Furuichi, S.** (2006). “Information theoretical properties of Tsallis entropies.” *Journal of Mathematical Physics*, 47, 023302. doi: 10.1063/1.2165744
- **Anastasiadis, A.** (2012). “Entropy properties and multiple Tsallis distributions.” *Entropy*, 14, 174-176. doi: 10.3390/e14020174
- **Alomani, G., & Kayid, M.** (2023). “Further properties of Tsallis entropy and its application.” *Entropy*, 25(2), 199. doi: 10.3390/e25020199

Divergence & Diversity Measures

- **van Erven, T., & Harremoës, P.** (2014). “Rényi divergence and Kullback-Leibler divergence.” *IEEE Transactions on Information Theory*, 60(7), 3797-3820. doi: 10.1109/TIT.2014.2320500
- **Jost, L.** (2006). “Entropy and diversity.” *Oikos*, 113(2), 363-375. doi: 10.1111/j.2006.0030-1299.14714.x
- **Chao, A., Chiu, C.-H., & Jost, L.** (2010). “Phylogenetic diversity measures based on Hill numbers.” *Philosophical Transactions of the Royal Society B*, 365(1558), 3599-3609. doi: 10.1098/rstb.2010.0272
- **Sason, I.** (2022). “Divergence measures: Mathematical foundations and applications in information-theoretic and statistical problems.” *Entropy*, 24(5), 712. doi: 10.3390/e24050712
- **Sfetcu, R.-C., Sfetcu, S.-C., & Preda, V.** (2022). “Some properties of weighted Tsallis and Kaniadakis divergences.” *Entropy*, 24(11), 1616. doi: 10.3390/e24111616

Applications in Biology

- **Yulmetyev, R. M., Emelyanova, N. A., & Gafarov, F. M.** (2004). “Dynamical Shannon entropy and information Tsallis entropy in complex systems.” *Physica A*, 341, 649-676. doi: 10.1016/j.physa.2004.03.094
 - **Baji'c, D.** (2022). “On quantization errors in approximate and sample entropy.” *Entropy*, 24(1), 73. doi: 10.3390/e24010073
 - **Baji'c, D.** (2024). “Information theory, living systems, and communication engineering.” *Entropy*, 26(5), 430. doi: 10.3390/e26050430
 - **Gao, X., Tsai, S.-B., Liu, F., Pan, L., & Deng, Y.** (2019). “Uncertainty measure based on Tsallis entropy in evidence theory.” *International Journal of Intelligent Systems*, 34(6), 1626-1647. doi: 10.1002/int.22185
-

Statistical Methods & Hypothesis Testing

Multiple Testing Correction

- **Benjamini, Y., & Hochberg, Y.** (1995). “Controlling the false discovery rate: a practical and powerful approach to multiple testing.” *Journal of the Royal Statistical Society B*, 57(1), 289-300.
- **Holm, S.** (1979). “A simple sequentially rejective multiple test procedure.” *Scandinavian Journal of Statistics*, 6, 65-70.

Permutation & Non-Parametric Methods

- **Ernst, M. D.** (2004). “Permutation methods: A basis for exact inference.” *Statistical Science*, 19(4), 676-685.
- **Meinshausen, N., Maathuis, M. H., & Bühlmann, P.** (2012). “Asymptotic optimality of the Westfall-Young permutation procedure for multiple testing under dependence.” *The Annals of Statistics*, 39(6), 3369-3391. doi: 10.1214/11-AOS946
- **Saulsbury, J. G.** (2020). “Permutation tests for comparative data.” *bioRxiv*. doi: 10.1101/2020.10.24.325472
- **Elkin, L. A., & Kay, M.** (2023). “An aligned rank transform procedure for multifactor contrast tests.” *Journal of Statistical Software* (forthcoming).

Effect Size & Interpretation

- **Cohen, J.** (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Lawrence Erlbaum Associates.
 - **Kraemer, H. C., & Kupfer, D. J.** (2005). “Size of treatment effects and their importance to clinical research and practice.” *Biological Psychiatry*, 59(11), 990-996. doi: 10.1016/j.biopsych.2005.09.014
 - **Cliff, N. D.** (1993). *Dominance statistics: Ordinal analyses for ordinal data*. JASS Monograph Series.
 - **Kerby, D. S.** (2014). “The simple difference formula: An approach to teaching nonparametric correlation.” *Comprehensive Psychology*, 3, 11.IT.3.1.
-

Linear & Generalized Additive Models

Linear Models & Mixed Effects

- **Hoff, P. D.** (2023). *Classical theory of linear statistical models* (2nd ed.). Springer.
- **Demidenko, E.** (2013). *Mixed models: Theory and applications with R* (2nd ed.). Wiley Series in Probability and Statistics.

Generalized Additive Models

- **Chapman, P.** (2017). *Generalized additive models: An introduction with R* (4th ed.). CRC Press.
- **Goude, Y.** (2024). “Forecasting at EDF: Generalized additive models for time series.” EDF R&D, EDF Lab Saclay.
- **Correia, H. E., & Abebe, A.** (2021). “Regularised rank quasi-likelihood estimation for generalised additive models.” *Journal of Nonparametric Statistics*, 33(1), 101-117. doi: 10.1080/10485252.2021.1921176

Generalized Estimating Equations

- **Liang, K.-Y., & Zeger, S. L.** (1986). “Longitudinal data analysis using generalized linear models.” *Biometrika*, 73(1), 13-22.
- **Hardin, J. W., & Hilbe, J. M.** (2013). “Generalized estimating equations.” *Encyclopedia of Biostatistics*, 4, 2057-2066.
- **Pan, W.** (2001). “Akaike’s information criterion in generalized estimating equations.” *Biometrics*, 57, 120-125.

Correlated & Longitudinal Data

- **Song, P. X.** (2007). *Correlated data analysis: Modeling, analytics, and applications*. Springer Series in Statistics.

Bootstrap & Resampling Methods

- **Efron, B.** (2010). *Large-scale inference: Empirical Bayes methods for estimation, testing, and prediction*. Cambridge University Press.
- **Efron, B., & Tibshirani, R. J.** (1993). *An introduction to the bootstrap*. Chapman and Hall.
- **Jordá Muñoz, R.** (2018). *Métodos de remuestreo: Jackknife y Bootstrap*. Master’s thesis, Universidad de Murcia, Facultad de Matemáticas.

Bayesian Methods

- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman and Hall/CRC.
 - Erhard, F., Hense, B., Jafari, M., Siebourg-Polster, J., Dölken, L., & Zimmer, R. (2018). “Improved Ribo-seq puromycin target reliability using Bayesian nonparametrics.” *Bioinformatics*, 34(12), 2096-2102.
-

Time Series Analysis

- Chakraborty, T. (2019). *Introductory time series analysis*. Indian Statistical Institute, Kolkata.
 - Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice* (2nd ed.). Retrieved from <https://otexts.com/fpp2/>
 - Jose, J., & Lal, P. S. (2013). “Application of ARIMA(1,1,0) model for predicting time delay of search engine crawlers.” *Informatica Economică*, 17(4), 26-39. doi: 10.12948/issn14531305/17.4.2013.03
-

Transcriptomics & Gene Expression Applications

Single-Cell & Isoform Analysis

- Cao, J., Packer, J. S., Ramani, V., Cusanovich, D. A., Huynh, C., Daza, R., . . . , Shendure, J. (2017). “Comprehensive single-cell transcriptional profiling of a multicellular organism.” *Science*, 357, 661-667. doi: 10.1126/science.aam8940
- Gandrillon, O., Gaillard, M., Espinasse, T., Garnier, N. B., Dussiau, C., Kosmiker, O., & Sujobert, P. (2021). “Entropy as a measure of variability and stemness in single-cell transcriptomics.” *Entropy*, 21(5), 450.
- Seweryn, M. T., Pietrzak, M., & Ma, Q. (2020). “Application of information theoretical approaches to assess diversity and similarity in single-cell transcriptomics.” *Computational and Structural Biotechnology Journal*, 18, 1830-1837. doi: 10.1016/j.csbj.2020.05.006

Information Theory in Genomics

- Golomb, R., Yoles, M., Fishilevich, S., Cohen, B., Savariego Peled, S., Dahary, D., Gokhman, D., & Pilpel, Y. (2026). “An information content principle explains regulatory patterns of gene expression across human tissues.” *bioRxiv*. doi: 10.64898/2026.02.19.706555
- Chanda, P., Costa, E., Hu, J., Sukumar, S., Van Hemert, J., & Walia, R. (2020). “Information theory in computational biology: Where we stand today.” *Entropy*, 22(6), 627. doi: 10.3390/e22060627
- Adami, C. (2004). “Information theory in molecular biology.” *Physics of Life Reviews*, 1(1), 3-22. doi: 10.1016/j.plrev.2004.01.002
- Hinostroza Fuentes, V. G., Toledo Tan, M. J., Kapetanaki, M., Rashidi, P., Huang, K., & Benos, P. V. (2026). “Information-theoretic methods in spatial transcriptomics.” *TechRxiv*. doi: 10.36227/techrxiv.177155947

- **Bartal, A., & Jagodnik, K. M.** (2022). “Progress in and opportunities for applying information theory to computational biology and bioinformatics.” *Entropy*, 24(7), 925. doi: 10.3390/e24070925

Computational Approaches

- **Ramírez-Reyes, A., Hernández-Montoya, A. R., Herrera-Corral, G., & Domínguez-Jiménez, I.** (2016). “Determining the entropic index q of Tsallis entropy in images through redundancy.” *Entropy*, 18(8), 299. doi: 10.3390/e18080299
 - **Ré, M. A., & Azad, R. K.** (2014). “Generalization of entropy based divergence measures for symbolic sequence analysis.” *PLoS ONE*, 9(4), e93532. doi: 10.1371/journal.pone.0093532
-

Cancer & Biological Systems

- **Tarabichi, M., Antoniou, A., Saiselet, M., Pita, J. M., Andry, G., Dumont, J. E., Detours, V., & Maenhaut, C.** (2013). “Systems biology of cancer: entropy, disorder, and selection-driven evolution to independence, invasion and ‘swarm intelligence’.” *Cancer Metastasis Reviews*, 32, 403-421. doi: 10.1007/s10555-013-9431-y
 - **Nijman, S. M. B.** (2020). “Perturbation-driven entropy as a source of cancer cell heterogeneity.” *Trends in Cancer*, 6(6), 454-462. doi: 10.1016/j.trecan.2020.02.016
-

Complexity & Entropy in Physics

- **Shiner, J. S., Davison, M., & Landsberg, P. T.** (2007). “Simple measures of complexity.” *Physica A*, 386, 101-118. doi: 10.1016/j.physa.2007.05.065
 - **Shiner, J. S., & others.** (2002). *Entropy and entropy generation: Fundamentals and applications*. Kluwer Academic Publishers.
-

Machine Learning & Control

- **Wang, Z., So, O., Gibson, J., Vlahov, B., Gandhi, M. S., Liu, G.-H., & Theodorou, E. A.** (2021). “Variational inference model predictive control using Tsallis divergence.” *arXiv preprint arXiv:2104.00241*. doi: 10.48550/arXiv.2104.00241
 - **Chernyshov, K.** (2009). “Tsallis divergence of order $1/2$ in system identification related problems.” Technical report, V. A. Trapeznikov Institute of Control Sciences, Moscow, Russia.
-

Validation & Quality Control

- **Tian, L., Dong, X., Sotero, S., Lucks, J. B., & Bustamante, C. D.** (2019). “Benchmarking null models of transcription.” *Genome Biology*, 20, 104.
-

Documentation & Best Practices

- **Xie, Y., Allaire, J. J., & Grolemund, G.** (2021). *R markdown: The definitive guide* (2nd ed.). CRC Press/Routledge.
 - **Zhu, H.** (2023). “Create awesome HTML table with kableExtra.” CRAN. Retrieved from <https://haozhu233.github.io/kableExtra/>
-

How to Cite TSENAT

```
@Manual{TSENAT2026,  
  title = {TSENAT: Tsallis Entropy Analysis Toolbox},  
  author = {Cristobal Gallardo},  
  year = {2026},  
  note = {R package version 0.99.0},  
  url = {https://github.com/gallardoalba/TSENAT}  
}
```

Or in text: Gallardo, C. (2026). TSENAT: Tsallis Entropy Analysis Toolbox. R package version 0.99.0. <https://github.com/gallardoalba/TSENAT>